



University of Zurich^{UZH}

High Performance Computing Lecture 12

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Cloud Computing



Google Cloud



Azure

openstack®



Containers in HPC



Singularity



Anything as a Service (XaaS)

Infrastructure as a Service (IaaS)

- Data Centres (cooled rooms) are expensive to build and operate
- There is economy of scale
- Data centres can be large and remote
- Servers are “Virtualized”

Containers as a Service (Caas)

- The same principle, but for containers
- What is a container? More later.

Cloud Computing



Google Cloud



Azure

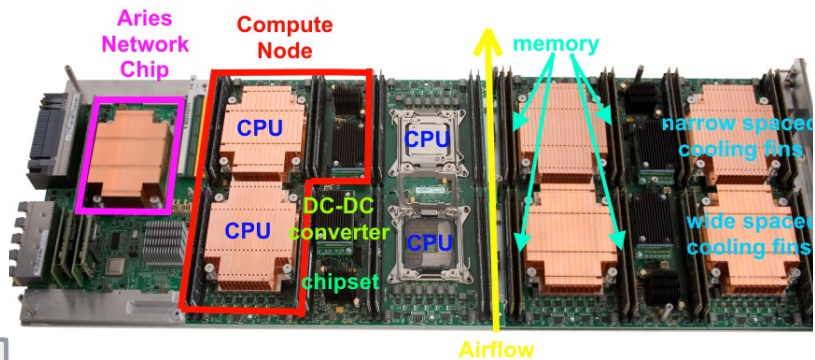
openstack®



Cloud
Computing



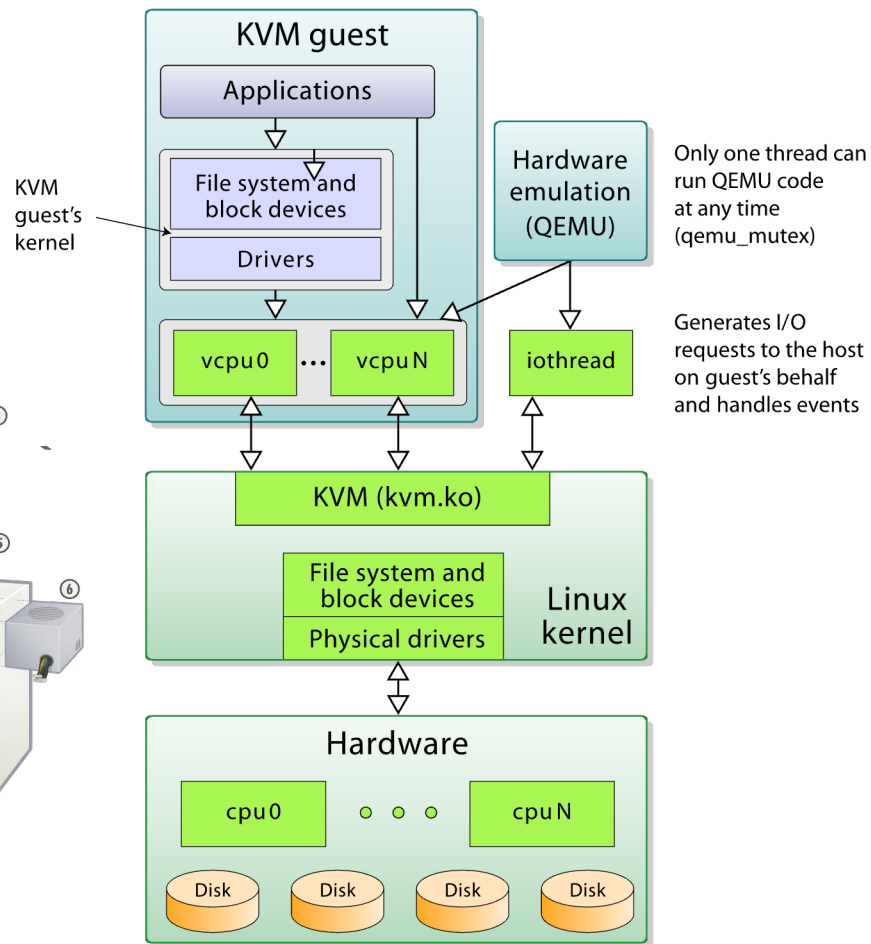
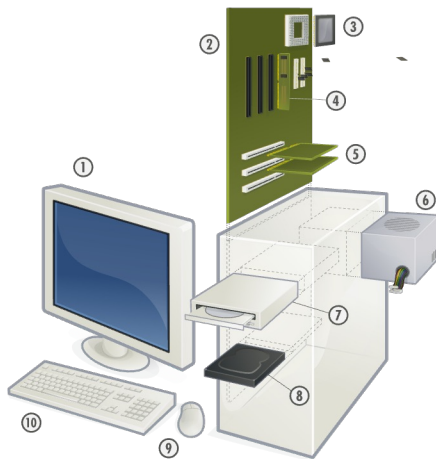
High Performance Computing



Hardware Components



What is a Virtual Machine (VM)





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Login



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UZH Cloud production environment

Log in

User Name

Password

Domain

Sign In

[User Documentation](#) - [About Science IT](#) - [ScienceCloud Support](#)

Status information

- HPCv4 VM flavors are currently unavailable. HPCv3 flavors remain available.

High Performance Computing

Overview

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Object Store

Identity

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dpotte

Project / Compute / Overview

Overview

Limit Summary

Compute

Instances

Used 0 of 50

VCPU

Used 0 of 400

RAM

Used 0B of 2TB

Volume

Volumes

Used 0 of 10

Volume Snapshots

Used 0 of 10

Volume Storage

Used 0B of 1000GB

Network

Floating IPs

Allocated 0 of 50

Security Groups

Used 1 of 10

Security Group Rules

Used 6 of 100

Networks

Used 0 of 100

Ports

Used 0 of 500

Routers

Used 0 of 10

2025-12-02

to

2025-12-03

Submit

Active Instances:

0

Active RAM:

0B

This Period's VCPU-Hours:

0.00

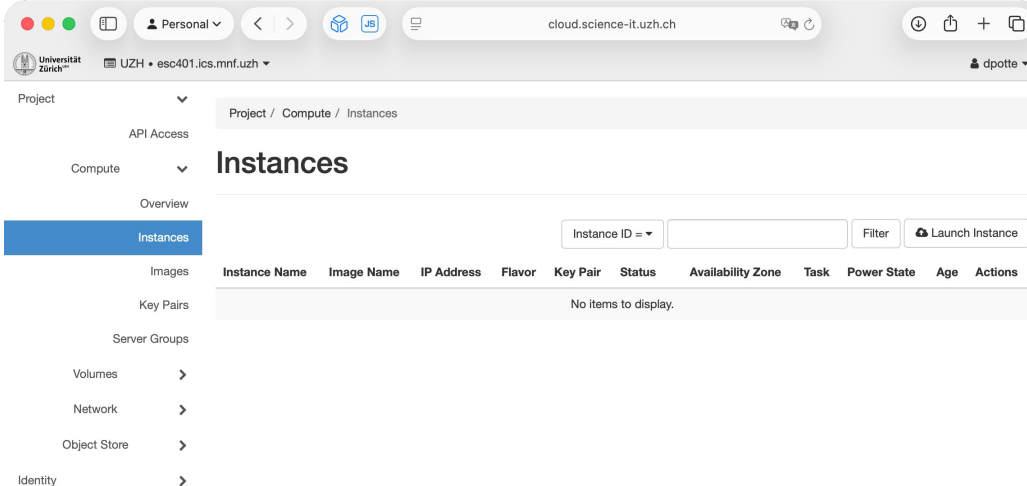
This Period's GB-Hours:

0.00

This Period's RAM-Hours:

0.00

Instances



The screenshot shows a web browser window with the URL `cloud.science-it.uzh.ch`. The page is titled "Instances" and is part of a navigation menu on the left. The menu includes "Project", "API Access", "Compute", "Overview", "Instances" (highlighted), "Images", "Key Pairs", "Server Groups", "Volumes", "Network", "Object Store", and "Identity". The main content area shows a breadcrumb "Project / Compute / Instances" and a table with columns: "Instance Name", "Image Name", "IP Address", "Flavor", "Key Pair", "Status", "Availability Zone", "Task", "Power State", "Age", and "Actions". The table is currently empty, displaying "No items to display." There is a search bar for "Instance ID" and a "Launch Instance" button.

Project / Compute / Instances

Instances

Instance ID =

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
No items to display.										



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Instance Name

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Launch Instance

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Details

Source *

Flavour *

Networks *

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Project Name

esc401.ics.mnf.uzh

Instance Name *

dpotter_test

Description

Test image for ESC #01

Availability Zone

nova

Count *

1

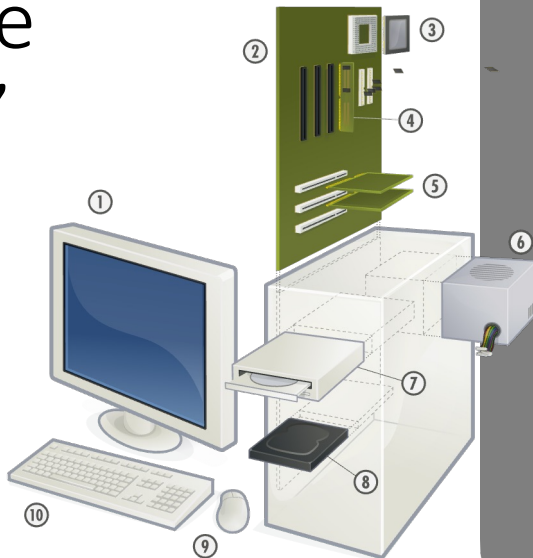
Total Instances (50 Max)

2%

0 Current Usage
1 Added
49 Remaining

Cancel Back Next Launch Instance

Source “Disk”



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Project

Launch Instance

Details

Source *

Flavour *

Networks *

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source

Image

Create New Volume

Yes No

Allocated

Displaying 0 items

Name	Updated	Size	Format	Visibility
Select an item from Available items below				

Displaying 0 items

Available 12

Select one

Q ubuntu X

Displaying 6 items

Name	Updated	Size	Format	Visibility
> ***CUDA+Singularity on Ubuntu 24.04 (2024-11-21) - L4 ONLY	9/11/25 9:18 AM	100.00 GB	RAW	Public
> ***Singularity on Ubuntu 24.04 (2024-11-21)	11/25/24 10:15 AM	100.00 GB	RAW	Public
> ***Ubuntu 22.04 (2024-11-13)	11/25/24 10:15 AM	100.00 GB	RAW	Public
> ***Ubuntu 24.04 (2024-11-13)	11/25/24 10:15 AM	100.00 GB	RAW	Public
> ubuntu-focal-20.04	12/1/25 12:50 PM	612.44 MB	QCOW2	Public
> ubuntu-jammy-22.04	12/1/25 12:51 PM	618.25 MB	QCOW2	Public

Displaying 6 items

X Cancel

< Back Next >

Launch Instance



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Source “Ubuntu”

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Launch Instance

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source **Create New Volume**

Image

Allocated

Displaying 1 item

Name	Updated	Size	Format	Visibility
> ***Ubuntu 24.04 (2024-11-13)	11/25/24 10:15 AM	100.00 GB	RAW	Public

Displaying 1 item

Available (11) Select one

Q ubuntu

Displaying 5 items

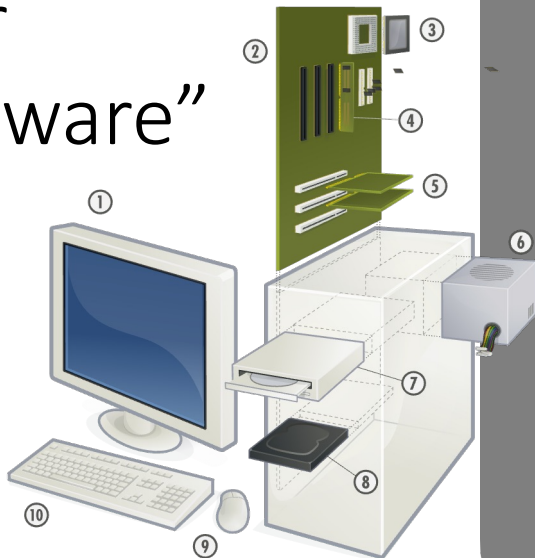
Name	Updated	Size	Format	Visibility
> ***CUDA+Singularity on Ubuntu 24.04 (2024-11-21) - L4 ONLY	9/11/25 9:18 AM	100.00 GB	RAW	Public
> ***Singularity on Ubuntu 24.04 (2024-11-21)	11/25/24 10:15 AM	100.00 GB	RAW	Public
> ***Ubuntu 22.04 (2024-11-13)	11/25/24 10:15 AM	100.00 GB	RAW	Public
> ubuntu-focal-20.04	12/1/25 12:50 PM	612.44 MB	QCOW2	Public
> ubuntu-jammy-22.04	12/1/25 12:51 PM	618.25 MB	QCOW2	Public

Displaying 5 items



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Flavor “Hardware”



Launch Instance

Flavours manage the sizing for the compute, memory and storage capacity of the instance.

Allocated
Displaying 0 items

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
Select a flavour from the available flavours below.						

Displaying 0 items

Available (20) Select one

Q hpcv3

Displaying 9 items

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
> 1cpu-4ram-hpcv3	1	4 GB	100 GB	100 GB	0 GB	Yes
> 2cpu-8ram-hpcv3	2	8 GB	100 GB	100 GB	0 GB	Yes
> 4cpu-16ram-hpcv3	4	16 GB	100 GB	100 GB	0 GB	Yes
> 8cpu-32ram-hpcv3	8	32 GB	100 GB	100 GB	0 GB	Yes
> 8cpu-32ram-hpcv3-gpuT4	8	32 GB	100 GB	100 GB	0 GB	Yes
> 15cpu-60ram-hpcv3-gpuT4	15	60 GB	100 GB	100 GB	0 GB	Yes
> 15cpu-60ram-hpcv3	15	60 GB	100 GB	100 GB	0 GB	Yes
> 30cpu-120ram-hpcv3	30	120 GB	100 GB	100 GB	0 GB	Yes
> 30cpu-120ram-hpcv3-gpuT4	30	120 GB	100 GB	100 GB	0 GB	Yes

Displaying 9 items

Cancel < Back Next > Launch Instance



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Flavor
2 cores
8 GB RAM

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Launch Instance

Flavours manage the sizing for the compute, memory and storage capacity of the instance.

Allocated
Displaying 1 item

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
> 2cpu-8ram-hpcv3	2	8 GB	100 GB	100 GB	0 GB	Yes

Displaying 1 item

Available (19) Select one

Q hpcv3 X

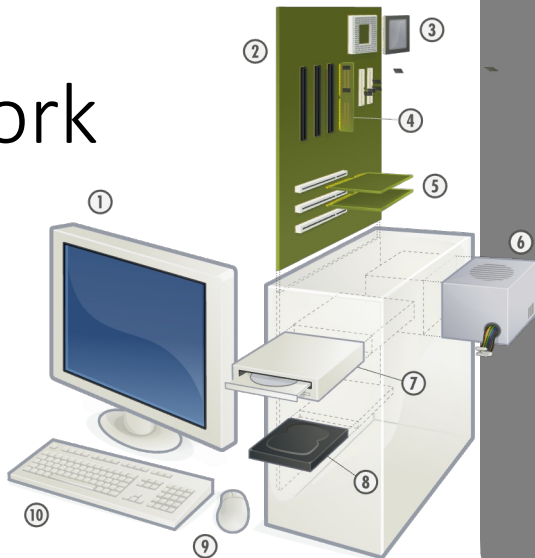
Displaying 8 items

Name	VCPUS	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
> 1cpu-4ram-hpcv3	1	4 GB	100 GB	100 GB	0 GB	Yes
> 4cpu-16ram-hpcv3	4	16 GB	100 GB	100 GB	0 GB	Yes
> 8cpu-32ram-hpcv3	8	32 GB	100 GB	100 GB	0 GB	Yes
> 8cpu-32ram-hpcv3-gpuT4	8	32 GB	100 GB	100 GB	0 GB	Yes
> 15cpu-60ram-hpcv3-gpuT4	15	60 GB	100 GB	100 GB	0 GB	Yes
> 15cpu-60ram-hpcv3	15	60 GB	100 GB	100 GB	0 GB	Yes
> 30cpu-120ram-hpcv3	30	120 GB	100 GB	100 GB	0 GB	Yes
> 30cpu-120ram-hpcv3-gpuT4	30	120 GB	100 GB	100 GB	0 GB	Yes

Displaying 8 items

X Cancel < Back Next > Launch Instance

Network



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Launch Instance

Details

Source

Flavour

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Networks provide communication channels for instances in the cloud. You can select ports instead of networks or a mix of both.

Allocated 1

Displaying 1 item

Network	Subnets Associated	Shared	Admin State	Status
> uzh-only	uzh-only	Yes	Up	Active

Displaying 1 item

Available 2

Select one or more

Click here for filters or full text search.

Displaying 2 items

Network	Subnets Associated	Shared	Admin State	Status
> External Network VLAN 9	ExternalNetwork	Yes	Up	Active
> External Network VLAN 24	ExternalNetwork24	Yes	Up	Active

Displaying 2 items

Cancel

< Back

Next >

Launch Instance



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Key Pair

The screenshot shows the 'Launch Instance' dialog in the AWS Management Console. The 'Key Pair' tab is selected in the left-hand navigation menu. The main content area displays instructions on key pairs and provides buttons to '+ Create Key Pair' and 'Import Key Pair'. Below this, there are two sections: 'Allocated' and 'Available'. The 'Allocated' section shows one item with the name 'ed25519' and type 'ssh'. The 'Available' section is currently empty, showing 'Displaying 0 items'. At the bottom of the dialog, there are buttons for 'Cancel', 'Back', 'Next', and 'Launch Instance'.

Launch Instance

A key pair allows you to SSH into your newly created instance. You may select an existing key pair, import a key pair, or generate a new key pair.

[+ Create Key Pair](#) [Import Key Pair](#)

Allocated

Displaying 1 item

Name	Type
ed25519	ssh

Displaying 1 item

Available 0 Select one

Click here for filters or full text search.

Displaying 0 items

Name	Type
No items to display.	

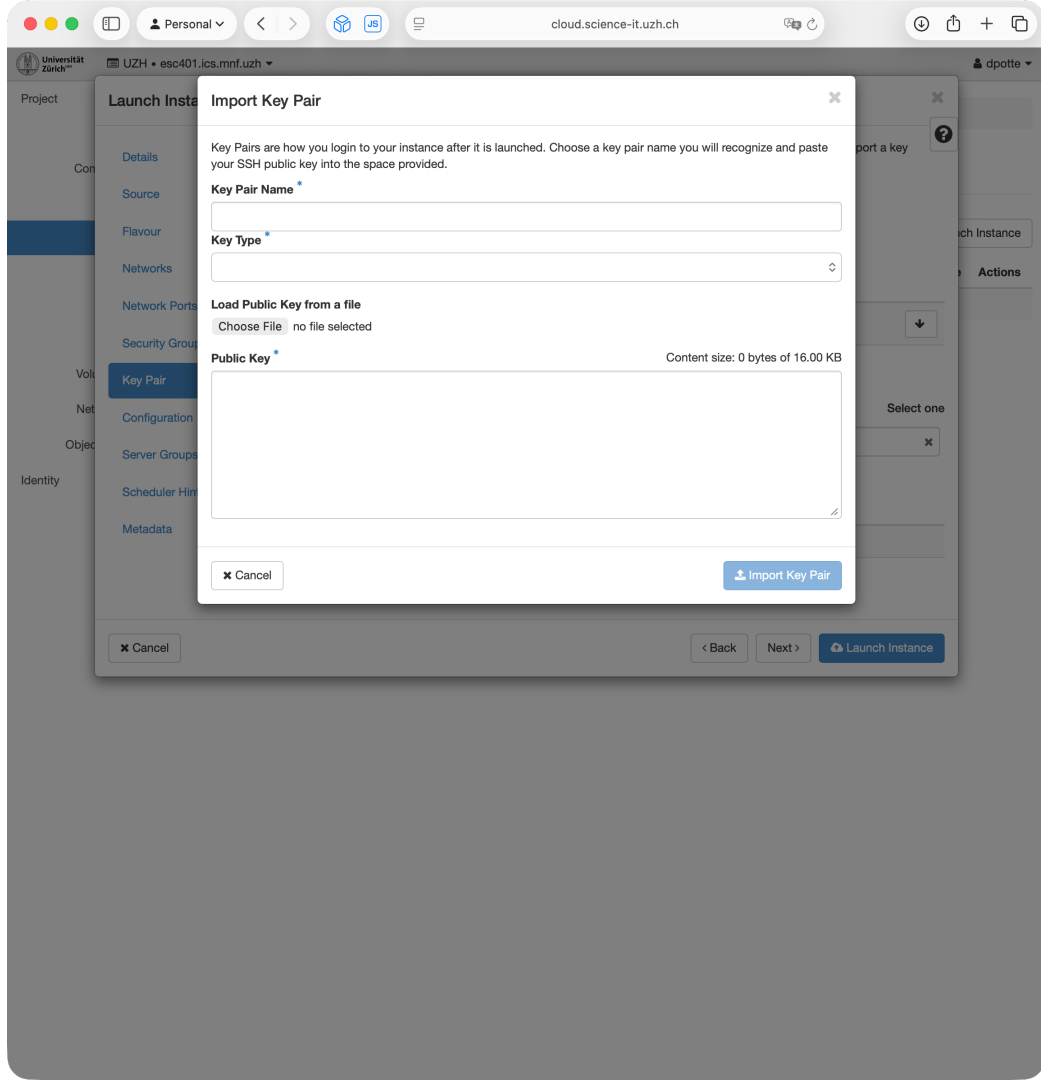
Displaying 0 items

[Cancel](#) [Back](#) [Next](#) [Launch Instance](#)

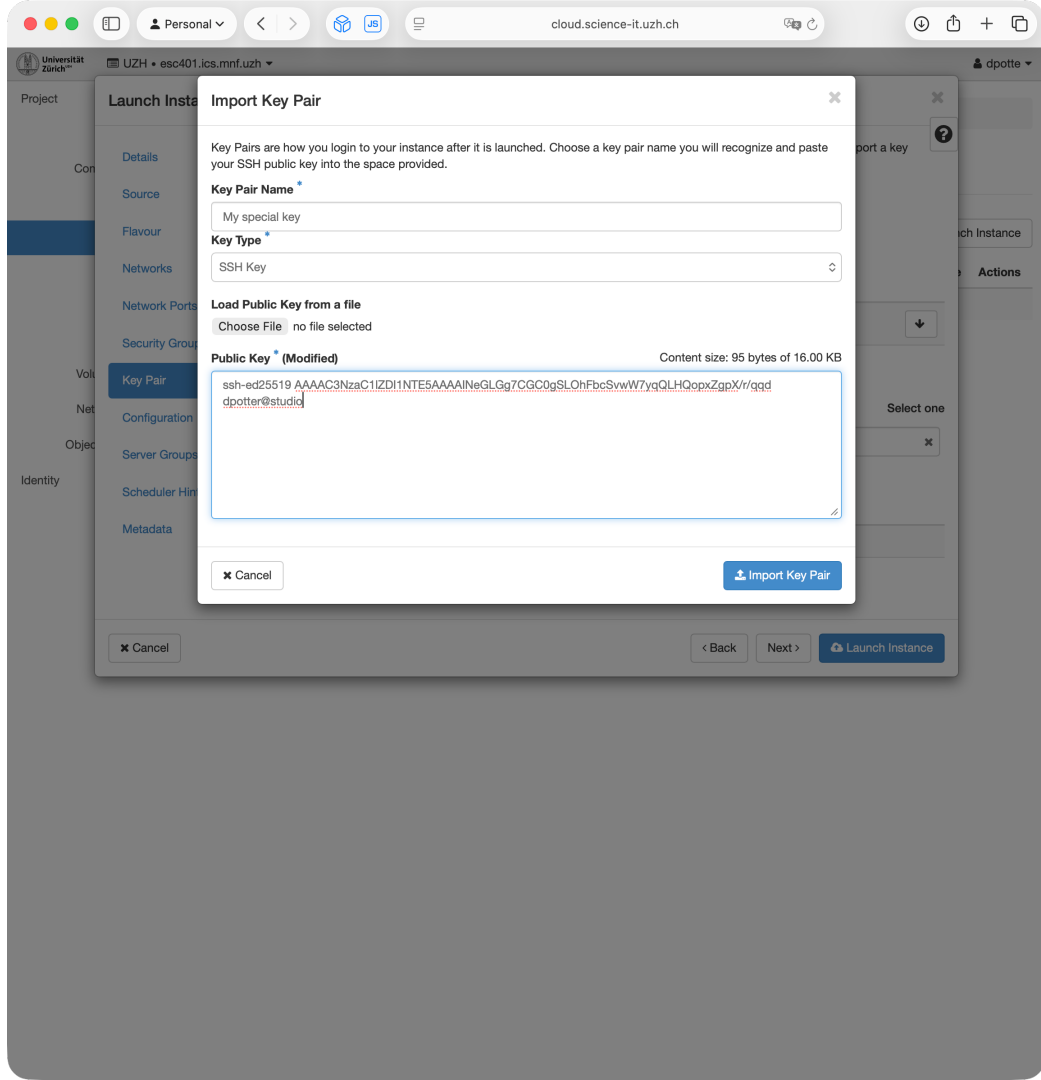


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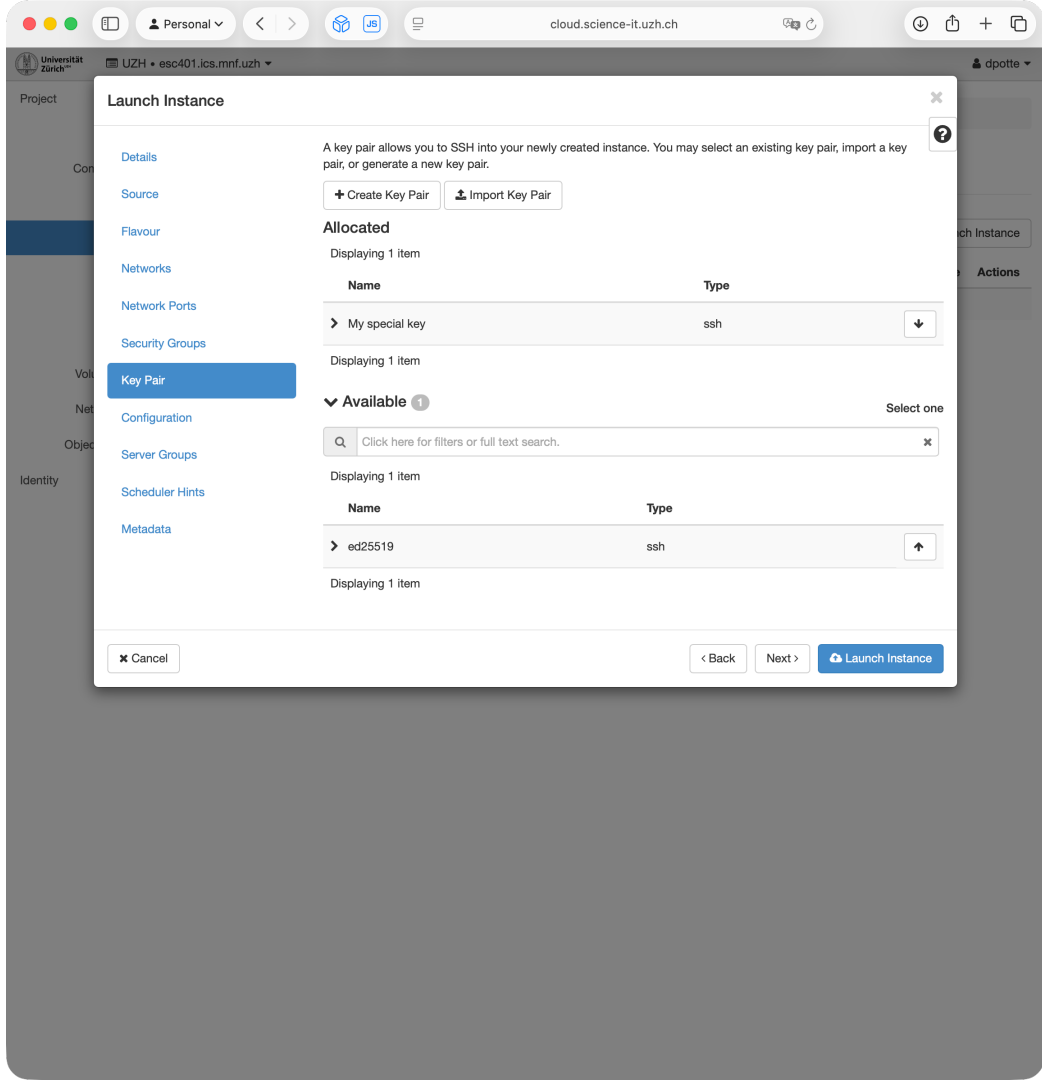
Import Key



Import Key



Import Key



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Launch Instance

A key pair allows you to SSH into your newly created instance. You may select an existing key pair, import a key pair, or generate a new key pair.

[+ Create Key Pair](#) [Import Key Pair](#)

Allocated

Displaying 1 item

Name	Type
> My special key	ssh

Displaying 1 item

Available 1

Select one

Click here for filters or full text search.

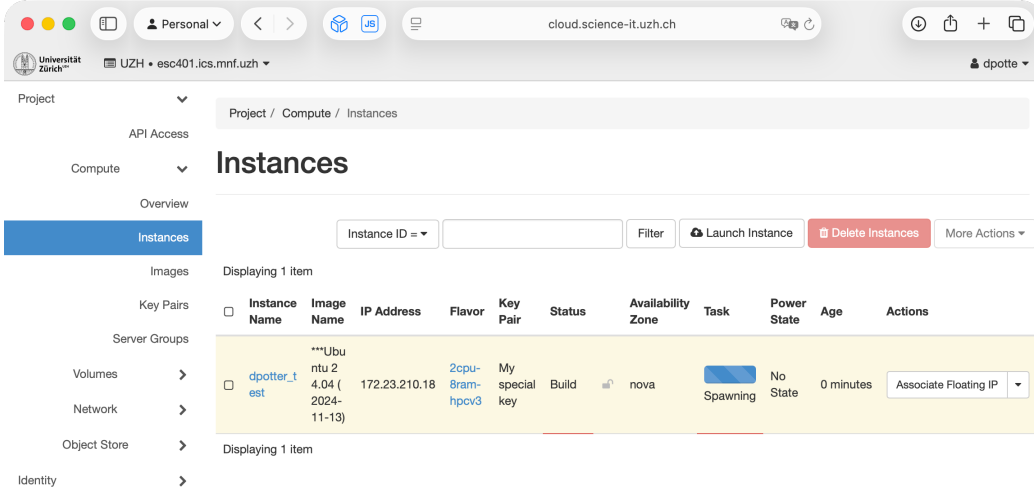
Displaying 1 item

Name	Type
> ed25519	ssh

Displaying 1 item

[Cancel](#) [Back](#) [Next](#) [Launch Instance](#)

Launching Instance



Project / Compute / Instances

Instances

Instance ID = Filter [Launch Instance](#) [Delete Instances](#) [More Actions](#)

Displaying 1 item

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	dpotter_test	***Ubuntu 20.04 (2024-11-13)	172.23.210.18	2cpu-8ram-hpcv3	My special key	Build	nova	Spawning	No State	0 minutes	Associate Floating IP

Displaying 1 item



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Launch Instance

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dpotte

Project / Compute / Instances

Instances

Instance ID = Filter [Launch Instance](#) [Delete Instances](#) [More Actions](#)

Displaying 1 item

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	dpotter_test	***Ubuntu 24.04 (2024-11-13)	172.23.210.18	2cpu-8ram-hpcv3	My special key	Active	nova	None	Running	0 minutes	Create Snapshot

Displaying 1 item

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Object Store

Identity



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Launch Instance

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Personal

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dpotte

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Object Store

Identity

Project / Compute / Instances

Instances

Instance ID = Filter [Launch Instance](#) [Delete Instances](#) [More Actions](#)

Displaying 1 item

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	dpotter_test	***Ubuntu 24.04 (2024-11-13)	172.23.210.18	2cpu-8ram-hpcv3	My special key	Active	nova	None	Running	0 minutes	Create Snapshot

Displaying 1 item

- Associate Floating IP
- Attach Interface
- Detach Interface
- Edit Instance
- Attach Volume
- Detach Volume
- Update Metadata
- Edit Security Groups
- Edit Port Security Groups
- Console
- View Log
- Rescue Instance
- Pause Instance
- Suspend Instance
- Shelve Instance
- Resize Instance
- Lock Instance
- Soft Reboot Instance
- Hard Reboot Instance
- Shut Off Instance
- Rebuild Instance
- Delete Instance



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Login!

(VPN needed)

```
ubuntu@dpotter-test: ~  
dpotter@studio ~ % ssh ubuntu@172.23.210.18  
The authenticity of host '172.23.210.18 (172.23.210.18)' can't be established.  
ED25519 key fingerprint is SHA256:2uXSIVuL/R1YUhhZ2L59pLnawNEXS1iFH+gdUGhAcI.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '172.23.210.18' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-48-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Wed Dec  3 09:36:49 UTC 2025  
  
System load:  0.06      Processes:            137  
Usage of /:   1.8% of 95.82GB  Users logged in:      0  
Memory usage: 2%        IPv4 address for enp3s0: 172.23.210.18  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
ubuntu@dpotter-test:~$ █
```



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VPN

Instructions

UZH - Knowhow Datenbank - E X

→ ↻ ↗

https://www.zi.uzh.ch/de/support/netzwerk/vpn.html

Google Suche

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Zentrale Informatik – Support

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Audio/Video-Infrastruktur

Beratung, Support, Schulung

E-Mail und Kollaboration

eLearning und Prüfungsunterstützung

Geschäftsanwendungen

Identität und Zugang

IT Arbeitsplatz

Kommunikation und Telefonie

Netzwerk

Externer Zugang zum Netz (VPN)

Mobilgeräte

Kabel-Verbindung (LAN)

Drahtlos-Verbindung (WLAN)

eduroam

Richtlinien, Anweisungen, Vorschriften

Visuelle Gestaltung und Video & Multimedia

Web- und Applikations-Hosting

Externer Zugang zum Netz (VPN)

Authentifizierung ist nicht gleich Verschlüsselung. An öffentlichen Netzwerken authentifizieren Sie sich zwar mit einem Passwort, der Datenverkehr verbleibt jedoch unverschlüsselt. Mittels einer "UZH Virtual Private Network (VPN)-Verbindung" werden öffentliche Verbindungen verschlüsselt, sodass UZH-Angehörige, ausserhalb der UZH-Gebäude, sicheren Zugriff auf das UZH-Netz haben, gerade so, als sässen sie mittendrin, gerade so, als würden sie sich direkt mit dem UZH-Netzwerk verbinden.

Für eine VPN-Verbindungsherstellung benötigen Sie keine zusätzliche Software, sondern lediglich eine bestehende Internet-Verbindung, sowie zusätzlich, wie nachfolgend schrittweise beschrieben, eine mit UZH-spezifischen (Verschlüsselungs-) Parametern konfigurierte aktive Netzwerk-Verbindung. Alternativ zur manuellen VPN-Konfiguration, stellt Ihnen die Zentrale Informatik hier ein Setup-Programm zur automatisierten VPN-Konfiguration unter Windows (Versionen 7 / 8 / 10) zur Verfügung.

show all

Windows - Automatische VPN-Konfiguration

1.) Konfigurationsdatei herunterladen

2.) Konfigurationsdatei ausführen

3.) VPN-Verbindung aufrufen

4.) VPN-Verbindung herstellen

... VPN-Verbindung entfernen

Windows - VPN-Konfiguration

1.) "Netzwerk- und Freigabecenter" in der Systemsteuerung aufrufen

2.) "Neue Verbindung oder neues Netzwerk einrichten" auswählen

3.) "Verbindung mit dem Arbeitsplatz herstellen" auswählen

4.) "Die Internetverbindung (VPN) verwenden" auswählen

5.) Internetadresse (Server) und Namen angeben

6.) Benutzerdaten angeben und Assistenten schliessen

7.) (Verbindungs-) Eigenschaften aufrufen

8.) VPN-Typ & Authentifizierung festlegen

9.) Erweiterte Eigenschaften angeben (Gruppenschlüssel)

10.) VPN-Verbindung aufrufen

11.) VPN-Verbindung herstellen

Mac OS X - VPN-Konfiguration

1.) Systemeinstellungen aufrufen

2.) Netzwerkeinstellungen öffnen

3.) Neue Verbindung erstellen



cowsay

```
ubuntu@dpotter-test:~$ cowsay
Command 'cowsay' not found, but can be installed with:
sudo apt install cowsay
ubuntu@dpotter-test:~$ sudo apt update -q
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://nova.clouds.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1625 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1340 kB]
Get:7 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [304 kB]
Get:8 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.7 kB]
Get:10 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1500 kB]
Get:11 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [304 kB]
Get:12 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:13 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.4 kB]
Get:14 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2307 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [219 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9448 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [915 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [206 kB]
Get:20 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [526 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.5 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.4 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2194 kB]
Get:24 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:25 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [516 B]
Get:26 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [30.3 kB]
Get:27 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [5808 B]
Get:28 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:29 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [484 B]
Get:30 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:31 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]
Get:32 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7128 B]
Get:33 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [501 kB]
Get:34 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:35 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [500 B]
Get:36 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [27.4 kB]
Get:37 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:38 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5956 B]
Get:39 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:40 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]
Get:41 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [29.2 kB]
Get:42 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [17.6 kB]
Get:43 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:44 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1444 B]
Get:45 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:46 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 13.2 MB in 2s (5349 kB/s)
Reading package lists...
Building dependency tree...
Reading state information...
255 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@dpotter-test:~$
```



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cowsay

```
ubuntu@dpotter-test:~$ sudo apt install -q cowsay
Reading package lists...
Building dependency tree...
Reading state information...
Suggested packages:
  filters cowsay-off
The following NEW packages will be installed:
  cowsay
0 upgraded, 1 newly installed, 0 to remove and 255 not upgraded.
Need to get 18.6 kB of archives.
After this operation, 93.2 kB of additional disk space will be used.
Get:1 http://nova.clouds.archive.ubuntu.com/ubuntu noble/universe amd64 cowsay all 3.03+dfsg2-8 [18.6 kB]
Fetched 18.6 kB in 0s (67.6 kB/s)
Selecting previously unselected package cowsay.
(Reading database ... 76155 files and directories currently installed.)
Preparing to unpack .../cowsay_3.03+dfsg2-8_all.deb ...
Unpacking cowsay (3.03+dfsg2-8) ...
Setting up cowsay (3.03+dfsg2-8) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@dpotter-test:~$ cowsay "Moo"
< Moo >
-----
      /\
     (oo)\_____
    (__)\       )\/\
        ||----w |
        ||     ||

ubuntu@dpotter-test:~$
```



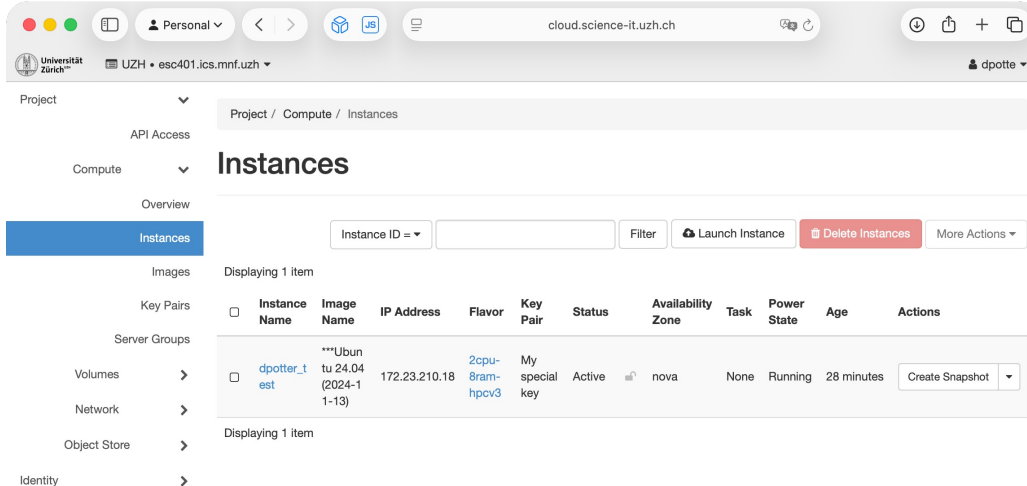

```
ubuntu@docker-test:~$ gcc
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
ubuntu@docker-test:~$ sudo apt install -q gcc cmake make
Reading package lists...
Building dependency tree...
Reading state information...
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu cmake-data cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu gcc-13 gcc-13-base
  gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libasan8 libatomic1 libbinutils libbct1-0 libctf-nobfd0 libctf0 libgcc-13-dev libgomp1
  libgprofng0 libhwasa0 libisl23 libitm1 libjsoncpp25 liblsan0 libmpc3 libquadmath0 librtls0 libstdc++11 libubsan1 libubsan1
Suggested packages:
  binutils-doc gprofng-gui cmake-doc cmake-format elpa-cmake-mode ninja-build cpp-doc gcc-13-locales cpp-13-doc gcc-multilib autoconf automake
  libtool flex bison gdb gcc-doc gcc-13-multilib gcc-13-doc gdb-x86-64-linux-gnu make-doc
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu cmake cmake-data cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu gcc gcc-13
  gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libasan8 libatomic1 libbinutils libbct1-0 libctf-nobfd0 libctf0 libgcc-13-dev
  libgomp1 libgprofng0 libhwasa0 libisl23 libitm1 libjsoncpp25 liblsan0 libmpc3 libquadmath0 librtls0 libstdc++11 libubsan1 libubsan1 make
0 upgraded, 35 newly installed, 0 to remove and 255 not upgraded.
Need to get 64.2 MB of archives.
After this operation, 217 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 binutils-common amd64 2.42-4ubuntu2.7 [240 kB]
Get:2 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libstdc++6 amd64 12.3.0-6ubuntu2-24.04 [10.7 MB]
Get:3 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libbinutils amd64 2.42-4ubuntu2.7 [577 kB]
Get:4 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libctf-nobfd0 amd64 2.42-4ubuntu2.7 [98.1 kB]
Get:5 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libctf0 amd64 2.42-4ubuntu2.7 [94.5 kB]
Get:6 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgprofng0 amd64 2.42-4ubuntu2.7 [849 kB]
Get:7 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 binutils-x86-64-linux-gnu amd64 2.42-4ubuntu2.7 [2463 kB]
Get:8 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 binutils amd64 2.42-4ubuntu2.7 [18.2 kB]
Get:9 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 libjsoncpp25 amd64 1.9.5-6build1 [82.8 kB]
Get:10 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 librtls0 amd64 1.4.3-3build1 [129 kB]
Get:11 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 cmake-data all 3.28.3-1build7 [2155 kB]
Get:12 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 cmake amd64 3.28.3-1build7 [11.2 MB]
Get:13 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-base amd64 13.3.0-6ubuntu2-24.04 [51.5 kB]
Get:14 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libisl23 amd64 0.26-3build1.1 [680 kB]
Get:15 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmpc3 amd64 1.3.1-1build1.1 [54.6 kB]
Get:16 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2-24.04 [10.7 MB]
Get:17 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13 amd64 13.3.0-6ubuntu2-24.04 [1038 B]
Get:18 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 cpp-x86-64-linux-gnu amd64 4:13.2.0-7ubuntu1 [5326 B]
Get:19 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 cpp amd64 4:13.2.0-7ubuntu1 [22.4 kB]
Get:20 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libbct1-0 amd64 14.2.0-4ubuntu2-24.04 [48.0 kB]
Get:21 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgomp1 amd64 14.2.0-4ubuntu2-24.04 [148 kB]
Get:22 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libitm1 amd64 14.2.0-4ubuntu2-24.04 [29.7 kB]
Get:23 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libatomic1 amd64 14.2.0-4ubuntu2-24.04 [10.5 kB]
Get:24 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libasan8 amd64 14.2.0-4ubuntu2-24.04 [3031 kB]
Get:25 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 liblsan0 amd64 14.2.0-4ubuntu2-24.04 [1322 kB]
Get:26 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 14.2.0-4ubuntu2-24.04 [2772 kB]
Get:27 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 14.2.0-4ubuntu2-24.04 [1184 kB]
Get:28 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libhwasa0 amd64 14.2.0-4ubuntu2-24.04 [1641 kB]
Get:29 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libquadmath0 amd64 14.2.0-4ubuntu2-24.04 [153 kB]
Get:30 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgcc-13-dev amd64 13.3.0-6ubuntu2-24.04 [2681 kB]
Get:31 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2-24.04 [21.1 MB]
Get:32 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13 amd64 13.3.0-6ubuntu2-24.04 [494 kB]
Get:33 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 gcc-x86-64-linux-gnu amd64 4:13.2.0-7ubuntu1 [1212 B]
Get:34 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 gcc amd64 4:13.2.0-7ubuntu1 [5018 B]
Get:35 http://nova.clouds.archive.ubuntu.com/ubuntu noble/main amd64 make amd64 4.3-4.1build2 [180 kB]
Fetched 64.2 MB in 2s (27.2 MB/s)
Extracting templates from packages: 100%
Selecting previously unselected package binutils-common:amd64.
```



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gcc

```
ubuntu@dpotter-test: ~  
Unpacking gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Selecting previously unselected package gcc.  
Preparing to unpack .../33-gcc-4%3a13.2.0-7ubuntu1 amd64.deb ...  
Unpacking gcc (4:13.2.0-7ubuntu1) ...  
Selecting previously unselected package make.  
Preparing to unpack .../34-make_4.3-4.1build2 amd64.deb ...  
Unpacking make (4.3-4.1build2) ...  
Setting up binutils-common:amd64 (2.42-4ubuntu2.7) ...  
Setting up libctf-nobfd0:amd64 (2.42-4ubuntu2.7) ...  
Setting up libgomp1:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up libsframe1:amd64 (2.42-4ubuntu2.7) ...  
Setting up gcc-13-base:amd64 (13.3.0-6ubuntu2~24.04) ...  
Setting up make (4.3-4.1build2) ...  
Setting up libquadmath0:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up libmpc3:amd64 (1.3.1-1build1.1) ...  
Setting up libatomic1:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up libjsongpp25:amd64 (1.9.5-6build1) ...  
Setting up libubsan1:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up libhwasan0:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up librtls0:amd64 (1.4.3-3build1) ...  
Setting up libasan8:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up cmake-data (3.28.3-1build7) ...  
Setting up libtsan2:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up libbinutils:amd64 (2.42-4ubuntu2.7) ...  
Setting up libisl23:amd64 (0.26-3build1.1) ...  
Setting up libcc1-0:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up liblsan0:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up libitm1:amd64 (14.2.0-4ubuntu2~24.04) ...  
Setting up libctf0:amd64 (2.42-4ubuntu2.7) ...  
Setting up cpp-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...  
Setting up libgprofng0:amd64 (2.42-4ubuntu2.7) ...  
Setting up libgcc-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...  
Setting up cmake (3.28.3-1build7) ...  
Setting up binutils-x86-64-linux-gnu (2.42-4ubuntu2.7) ...  
Setting up cpp-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Setting up cpp-13 (13.3.0-6ubuntu2~24.04) ...  
Setting up gcc-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...  
Setting up binutils (2.42-4ubuntu2.7) ...  
Setting up gcc-13 (13.3.0-6ubuntu2~24.04) ...  
Setting up cpp (4:13.2.0-7ubuntu1) ...  
Setting up gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Setting up gcc (4:13.2.0-7ubuntu1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@dpotter-test:~$ gcc  
gcc: fatal error: no input files  
compilation terminated.  
ubuntu@dpotter-test:~$
```

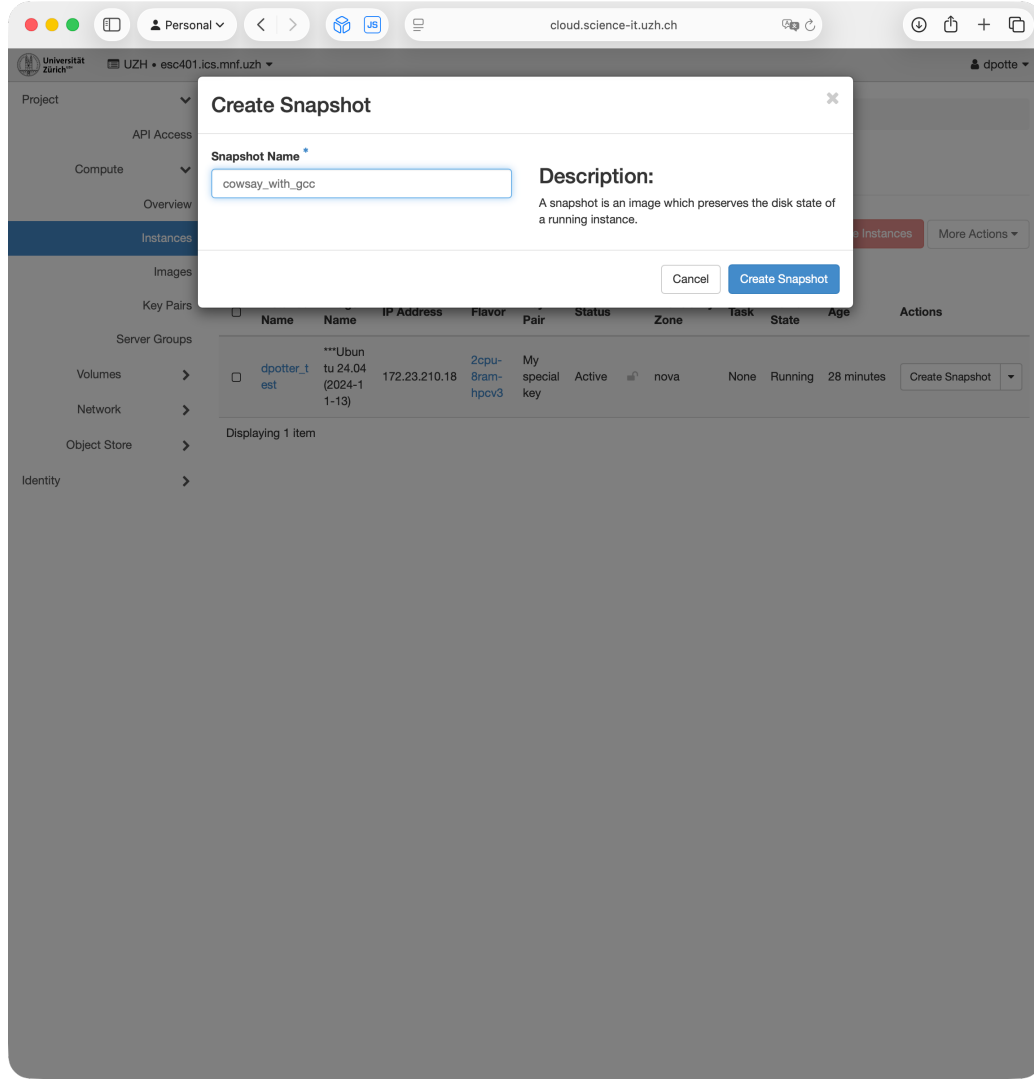


The screenshot shows a web browser window with the URL `cloud.science-it.uzh.ch`. The page is titled "Instances" and displays a table of running cloud instances. The left sidebar contains a navigation menu with options: Project, API Access, Compute, Overview, Instances (selected), Images, Key Pairs, Server Groups, Volumes, Network, Object Store, and Identity. The main content area shows a table with columns: Instance Name, Image Name, IP Address, Flavor, Key Pair, Status, Availability Zone, Task, Power State, Age, and Actions. One instance is listed: `dpotter_test`, using the `***Ubuntu 24.04 (2024-11-13)` image, with IP `172.23.210.18`, `2cpu-8ram-hpcv3` flavor, and `My special key` key pair. The instance is in an `Active` state in the `nova` availability zone, with a `None` task, `Running` power state, and an age of `28 minutes`. The `Actions` column for this instance includes a `Create Snapshot` button.

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
<code>dpotter_test</code>	<code>***Ubuntu 24.04 (2024-11-13)</code>	<code>172.23.210.18</code>	<code>2cpu-8ram-hpcv3</code>	<code>My special key</code>	Active	<code>nova</code>	None	Running	28 minutes	<button>Create Snapshot</button>

Snapshot

Snapshot



The screenshot shows a web browser window with the URL `cloud.science-it.uzh.ch`. The page displays the OpenStack dashboard interface. A modal dialog titled "Create Snapshot" is open, allowing the user to create a snapshot of a running instance.

Create Snapshot

Snapshot Name *

`cowsay_with_gcc`

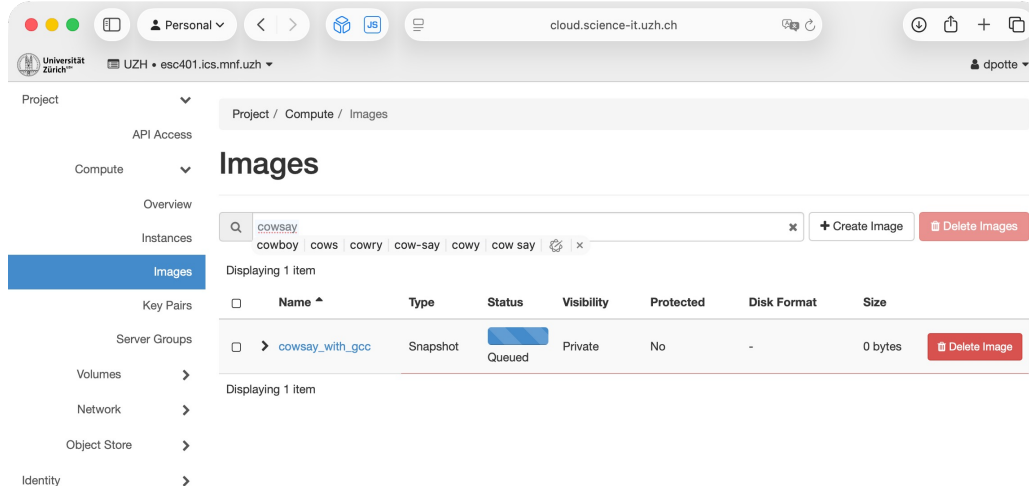
Description:

A snapshot is an image which preserves the disk state of a running instance.

The background shows the "Instances" section of the dashboard. A table lists the available instances:

Name	Name	IP Address	Flavor	Pair	Status	Zone	Task	State	Age	Actions		
☐	<code>dpotter_t est</code>	***Ubun tu 24.04 (2024-1 1-13)	172.23.210.18	2cpu- 8ram- hpcv3	My special key	Active	eu	nova	None	Running	28 minutes	Create Snapshot More Actions

Displaying 1 item



Project / Compute / Images

Images

Search: cowsay
cowsay cowboy cows cowry cow-say cowy cow say x

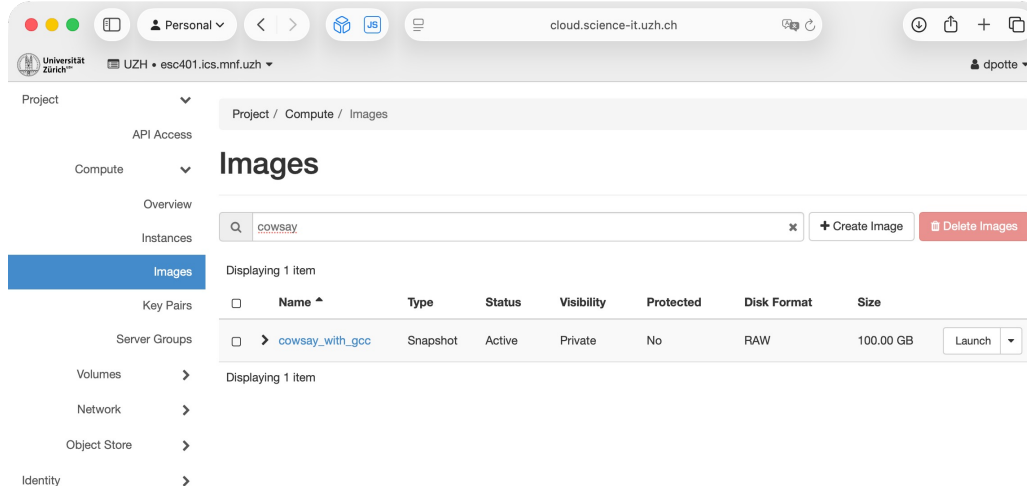
+ Create Image Delete Images

Displaying 1 item

☐	Name ^	Type	Status	Visibility	Protected	Disk Format	Size	
☐	> cowsay_with_gcc	Snapshot	Queued	Private	No	-	0 bytes	Delete Image

Displaying 1 item

Snapshot

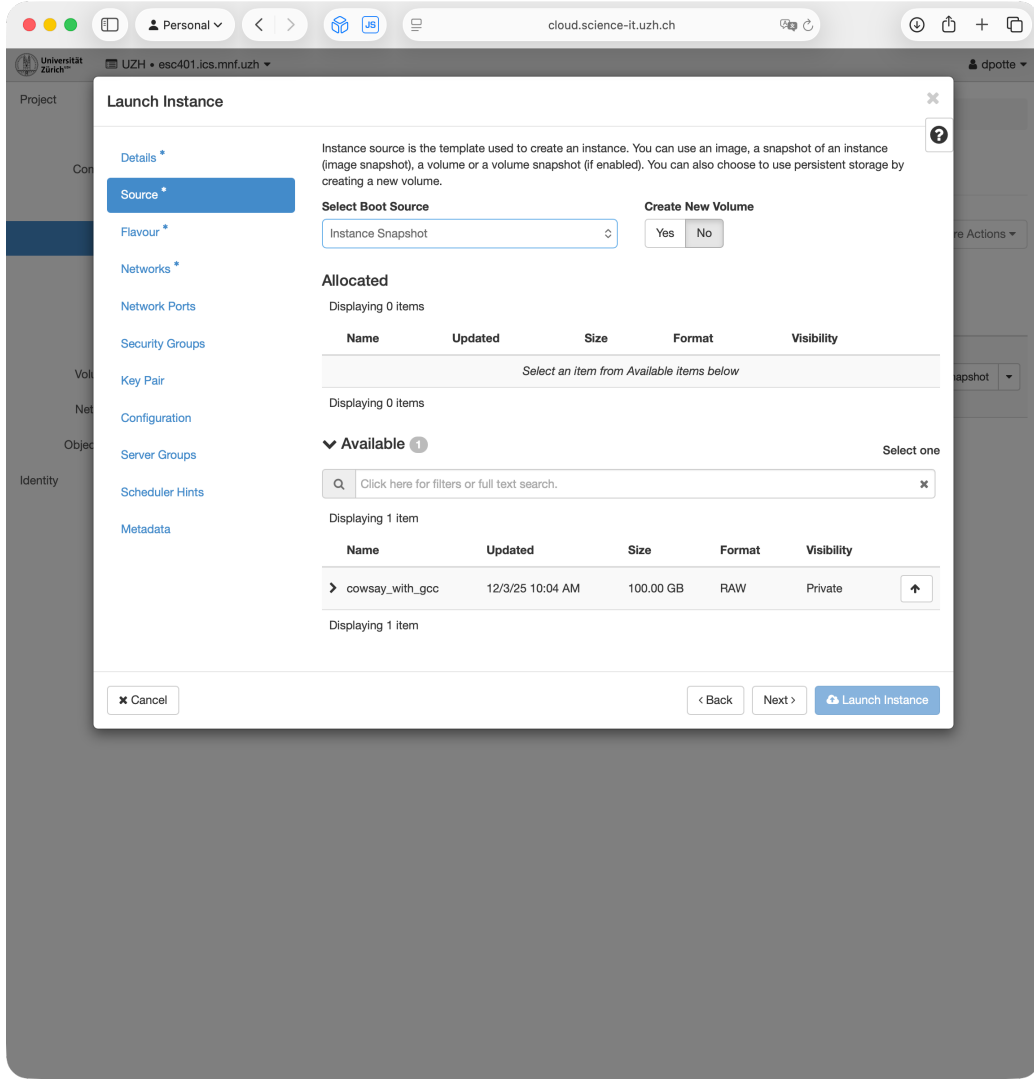


The screenshot shows a web browser window with the URL `cloud.science-it.uzh.ch`. The page displays a navigation sidebar on the left with options: Project, API Access, Compute, Overview, Instances, **Images** (selected), Key Pairs, Server Groups, Volumes, Network, Object Store, and Identity. The main content area is titled 'Images' and shows a search bar with the text 'cowsay'. Below the search bar, it says 'Displaying 1 item'. A table lists the image with the following columns: Name, Type, Status, Visibility, Protected, Disk Format, and Size. The table contains one entry: 'cowsay_with_gcc' (with a dropdown arrow next to the name), Type: Snapshot, Status: Active, Visibility: Private, Protected: No, Disk Format: RAW, and Size: 100.00 GB. There is a 'Launch' button and a dropdown arrow next to the image name. At the top right of the main content area, there are buttons for '+ Create Image' and 'Delete Images'.

Name ^	Type	Status	Visibility	Protected	Disk Format	Size
> cowsay_with_gcc	Snapshot	Active	Private	No	RAW	100.00 GB

Snapshot

Snapshot



The screenshot shows the 'Launch Instance' wizard in the AWS Management Console, specifically the 'Source' step. The left sidebar contains navigation links: Details, Source (selected), Networks, Network Ports, Security Groups, Key Pair, Configuration, Server Groups, Scheduler Hints, and Metadata. The main content area explains that the instance source is the template used to create an instance, and it can be an image, snapshot, or volume snapshot. It also mentions the option to create a new volume. Below this, there are two sections: 'Allocated' and 'Available'. The 'Allocated' section shows 'Displaying 0 items'. The 'Available' section shows 'Displaying 1 item' and includes a search bar. The available item is 'cowsay_with_gcc' with a size of 100.00 GB, format of RAW, and visibility of Private. At the bottom, there are buttons for 'Cancel', 'Back', 'Next', and 'Launch Instance'.

Launch Instance

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source

Instance Snapshot

Create New Volume

Yes No

Allocated

Displaying 0 items

Name	Updated	Size	Format	Visibility
Select an item from Available items below				

Displaying 0 items

Available 1

Select one

Click here for filters or full text search.

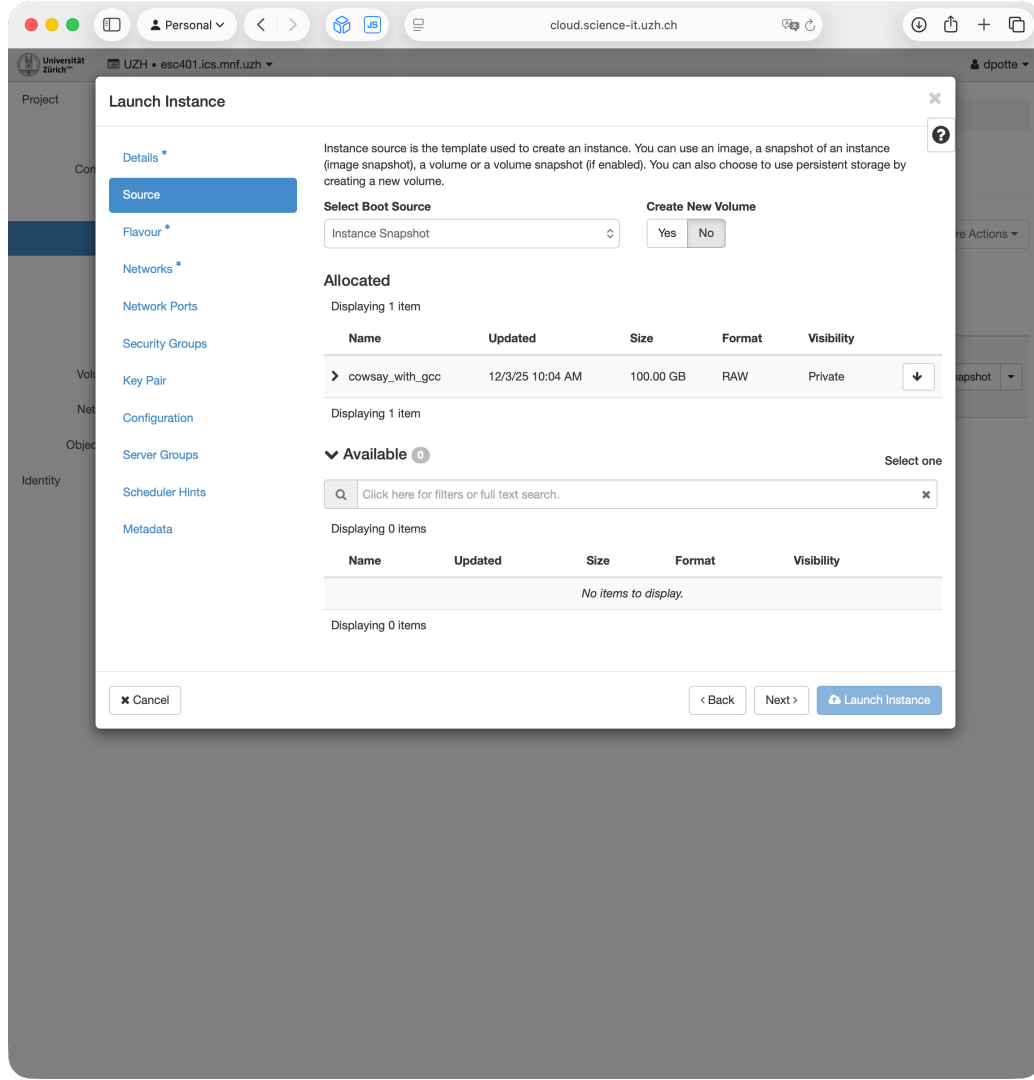
Displaying 1 item

Name	Updated	Size	Format	Visibility
cowsay_with_gcc	12/3/25 10:04 AM	100.00 GB	RAW	Private

Displaying 1 item

Cancel Back Next Launch Instance

Snapshot



Launch Instance

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source

Instance Snapshot

Create New Volume

Yes No

Allocated

Displaying 1 item

Name	Updated	Size	Format	Visibility
> cowsay_with_gcc	12/3/25 10:04 AM	100.00 GB	RAW	Private

Displaying 1 item

Available 0

Select one

Click here for filters or full text search.

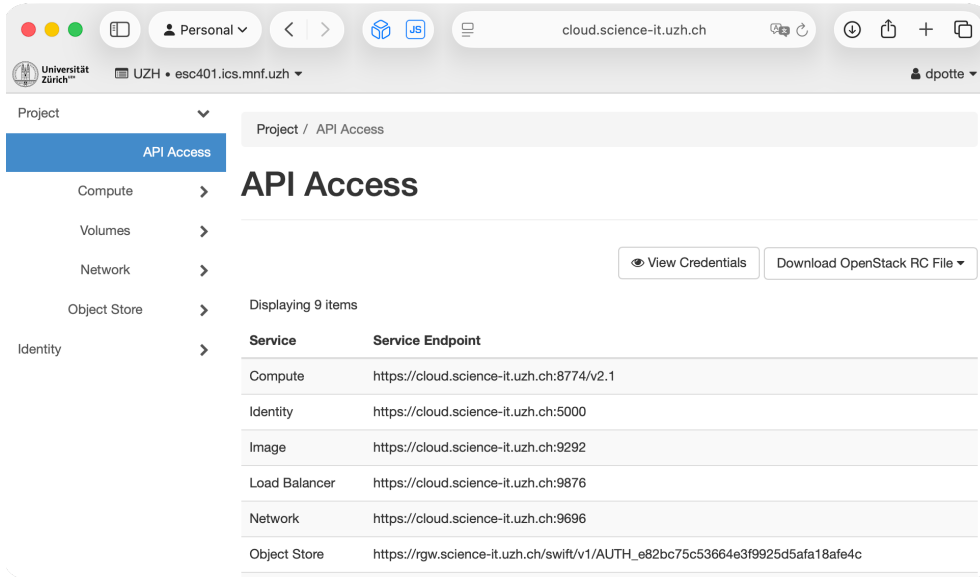
Displaying 0 items

Name	Updated	Size	Format	Visibility
No items to display.				

Displaying 0 items

Cancel Back Next Launch Instance

Command Line



The screenshot shows a web browser window with the URL `cloud.science-it.uzh.ch`. The page is titled "API Access" and displays a table of service endpoints. The left sidebar shows a navigation menu with "API Access" selected. The table lists services and their corresponding endpoints.

Service	Service Endpoint
Compute	<code>https://cloud.science-it.uzh.ch:8774/v2.1</code>
Identity	<code>https://cloud.science-it.uzh.ch:5000</code>
Image	<code>https://cloud.science-it.uzh.ch:9292</code>
Load Balancer	<code>https://cloud.science-it.uzh.ch:9876</code>
Network	<code>https://cloud.science-it.uzh.ch:9696</code>
Object Store	<code>https://rgw.science-it.uzh.ch/swift/v1/AUTH_e82bc75c53664e3f9925d5afa18afe4c</code>

- Access & Security Tab
- Download Openstack RC File v3 (version 3)



Using

```
#!/usr/bin/env bash

# To use an OpenStack cloud you need to authenticate against the Identity
# service named keystone, which returns a **Token** and **Service Catalog**.
# The catalog contains the endpoints for all services the user/tenant has
# access to - such as Compute, Image Service, Identity, Object Storage, Block
# Storage, and Networking (code-named nova, glance, keystone, swift,
# cinder, and neutron).
#
# *NOTE*: Using the 3 *Identity API* does not necessarily mean any other
# OpenStack API is version 3. For example, your cloud provider may implement
# Image API v1.1, Block Storage API v2, and Compute API v2.0. OS_AUTH_URL is
# only for the Identity API served through keystone.
export OS_AUTH_URL=https://cloud.s3it.uzh.ch:5000/v3

# With the addition of Keystone we have standardized on the term **project**
# as the entity that owns the resources.
export OS_PROJECT_ID=3bf33d6aadeb4b40a174576cbc30783f
export OS_PROJECT_NAME="esc401.ics.mnf.uzh"
export OS_USER_DOMAIN_NAME="Default"
if [ -z "$OS_USER_DOMAIN_NAME" ]; then unset OS_USER_DOMAIN_NAME; fi

# unset v2.0 items in case set
unset OS_TENANT_ID
unset OS_TENANT_NAME

# In addition to the owning entity (tenant), OpenStack stores the entity
# performing the action as the **user**.
export OS_USERNAME="dpotte"

# With Keystone you pass the keystone password.
echo "Please enter your OpenStack Password: "
read -sr OS_PASSWORD_INPUT
export OS_PASSWORD=$OS_PASSWORD_INPUT

# If your configuration has multiple regions, we set that information here.
# OS_REGION_NAME is optional and only valid in certain environments.
export OS_REGION_NAME="RegionOne"
# Don't leave a blank variable, unset it if it was empty
if [ -z "$OS_REGION_NAME" ]; then unset OS_REGION_NAME; fi

export OS_INTERFACE=public
export OS_IDENTITY_API_VERSION=3
```

douglass-macbook-pro:~\$ **source Downloads/esc401.ics.mnf.uzh-openrc.sh**

Please enter your OpenStack Password:

douglass-macbook-pro:~\$ **openstack flavor list -c ID -c Name**

ID	Name
b31d0cd7-96bd-4df7-ade5-4824754a4820	8cpu-32ram-hpcv3

douglass-macbook-pro:Cloud\$ **openstack image list -c ID -c Name | grep 20.04**

ID	Name
8b658463-195e-42d1-a718-becc647c5c1d	***CUDA latest on Ubuntu 20.04 (2021-03-19)
294ca836-5cf7-465c-8483-07e7e947f5eb	***Ubuntu 20.04 (2021-03-19)
b14e8803-e840-4a3b-bfa3-c3523c86c6fa	Ubuntu 20.04 (2020-11-18)



Scripts

```
$ cat make.sh
```

```
#!/bin/bash
```

```
NAME=dpotte_test
```

```
openstack server create --wait\
```

```
--flavor b31d0cd7-96bd-4df7-ade5-4824754a4820\
```

```
--image 294ca836-5cf7-465c-8483-07e7e947f5eb\
```

```
--key-name 'dpotter secure access' \
```

```
--security-group default\
```

```
--user-data user.sh\
```

```
$NAME
```

```
$ cat user.sh
```

```
#!/bin/bash
```

```
sudo apt update -q
```

```
sudo apt-get install -y\
```

```
  cowsay gcc cmake make
```



Create a server

```
douglass-macbook-pro:Cloud$ openstack server list
```

```
douglass-macbook-pro:Cloud$ ./make.sh
```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	nova
OS-EXT-STS:power_state	Running
OS-EXT-STS:task_state	None
OS-EXT-STS:vm_state	active
OS-SRV-USG:launched_at	2021-04-28T06:37:58.000000
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	uzh-only=172.23.8.51
adminPass	C3zwcC494Lnf
config drive	
created	2021-04-28T06:37:48Z
flavor	8cpu-32ram-hpcv3 (b31d0cd7-96bd-4df7-ade5-4824754a4820)
hostId	3a4e03111e94a485acdfe84c6519cb8547aea48137cbb59106782a54
id	b2c1ec9d-17de-4574-9e4b-ff65a54e7165
image	***Ubuntu 20.04 (2021-03-19) (294ca836-5cf7-465c-8483-07e7e947f5eb)
key name	dpotter secure access
name	dpotte_test
progress	0
project id	3bf33d6aadeb4b40a174576cbc30783f
properties	
security_groups	name='default'
status	ACTIVE
updated	2021-04-28T06:37:59Z
user id	dpotte
volumes_attached	

```
douglass-macbook-pro:Cloud$ openstack server list
```

ID	Name	Status	Networks	Image	Flavor
b2c1ec9d-17de-4574-9e4b-ff65a54e7165	dpotte_test	ACTIVE	uzh-only=172.23.8.51	***Ubuntu 20.04 (2021-03-19)	8cpu-32ram-hpcv3



Logging In

```
douglass-macbook-pro:Cloud$ ssh ubuntu@172.23.8.51
ssh: connect to host 172.23.8.51 port 22: Connection refused
douglass-macbook-pro:Cloud$ ssh ubuntu@172.23.8.51
The authenticity of host '172.23.8.51 (172.23.8.51)' can't be established.
ECDSA key fingerprint is SHA256:6VVoE6elUj2ArV2M/CEilovDT9/J7FtSPgJQMBBy2Tyk.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.23.8.51' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-67-generic x86_64)
```

```
...
Last login: Fri Mar 19 13:52:57 2021 from 172.23.255.1
ubuntu@dpotte-test:~$ cowsay "moo"
```

Command 'cowsay' not found, but can be installed with:

```
sudo apt install cowsay
```

```
ubuntu@dpotte-test:~$ cowsay "moo"
```

```
< moo >
-----
      /\
     (oo)\_____)
     ( )\      )\/\
        ||----w |
        ||     ||
```



Cleaning up

```
douglass-macbook-pro:Cloud$ openstack server list
```

ID	Name	Status	Networks	Image	Flavor
b2c1ec9d-17de-4574-9e4b-ff65a54e7165	dpotte_test	ACTIVE	uzh-only=172.23.8.51	***Ubuntu 20.04 (2021-03-19)	8cpu-32ram-hpcv3

```
douglass-macbook-pro:Cloud$ openstack server delete dpotte_test
```

```
douglass-macbook-pro:Cloud$ openstack server list
```

ID	Name	Status	Networks	Image	Flavor
b2c1ec9d-17de-4574-9e4b-ff65a54e7165	dpotte_test	ACTIVE	uzh-only=172.23.8.51	***Ubuntu 20.04 (2021-03-19)	8cpu-32ram-hpcv3

```
douglass-macbook-pro:Cloud$ openstack server list
```

```
douglass-macbook-pro:Cloud$
```

Summary

- Create an instance with the right “hardware”
- Log into our new instance (“Server”)
`ssh ubuntu@172.23.x.x`
- Update package catalogue:
`sudo apt update`
- Install package(s):
`sudo apt install cowsay`
`sudo apt install gcc cmake`
- Create a snapshot if needed
- Can be automated in BASH or Python



Omitted

- Attaching additional storage (“Volumes”)
 - Image changes disappear when instance deleted
 - Unless a snapshot is taken
 - Image is rather small – 100 GB maximum
- Using the API (in for example Python)
 - You can use Python instead of the web interface
 - Allows multiple instances to be launched
- Orchestration
 - Create a pipeline and execute it
 - Analogous to queue systems

Containers in HPC



Singularity



Containers in HPC



docker

SHIFTER

Singularity

Why Containers?

- We need a custom environment like a VM
 - Example: want to run an Ubuntu code on Piz Daint
- Software setup is difficult
 - Requires a specific operating system (e.g., Redhat)
 - Requires specific versions of libraries
 - Compilation process is difficult and time consuming
 - Binary only distribution with dependencies
- Solutions
 - Provide a VM image
 - Provide a container
- Containers are lightweight

TOOLS

Containers

Sarus

Shifter

Singularity

Continuous Integration

High Throughput Scheduler

Interactive Supercomputing

Julia

Python

JupyterLab

OpenStack

USEFUL LINKS

Account and Resources Tool

CSCS Website

Events

Tutorials

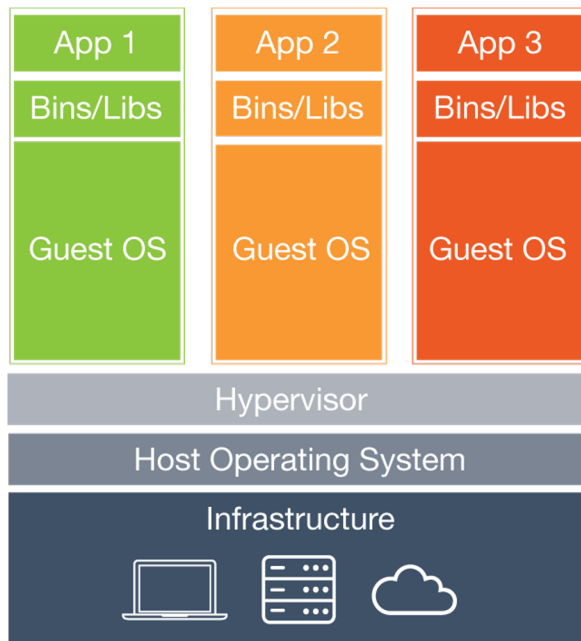
Production Repository

Sarus

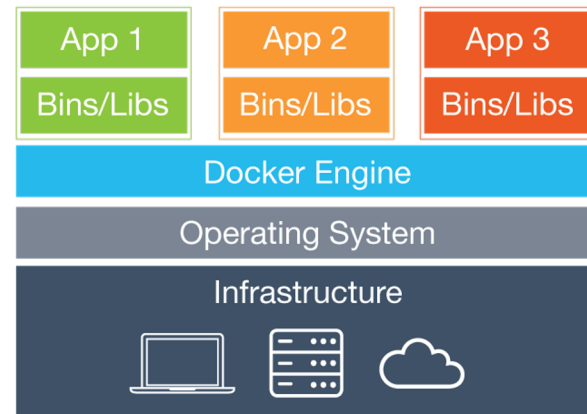
Sarus is a software to run Linux containers on High Performance Computing environments. Its development has been driven by the specific requirements of HPC systems, while leveraging open standards and technologies to encourage vendor and community involvement.



Container vs. Virtual Machine

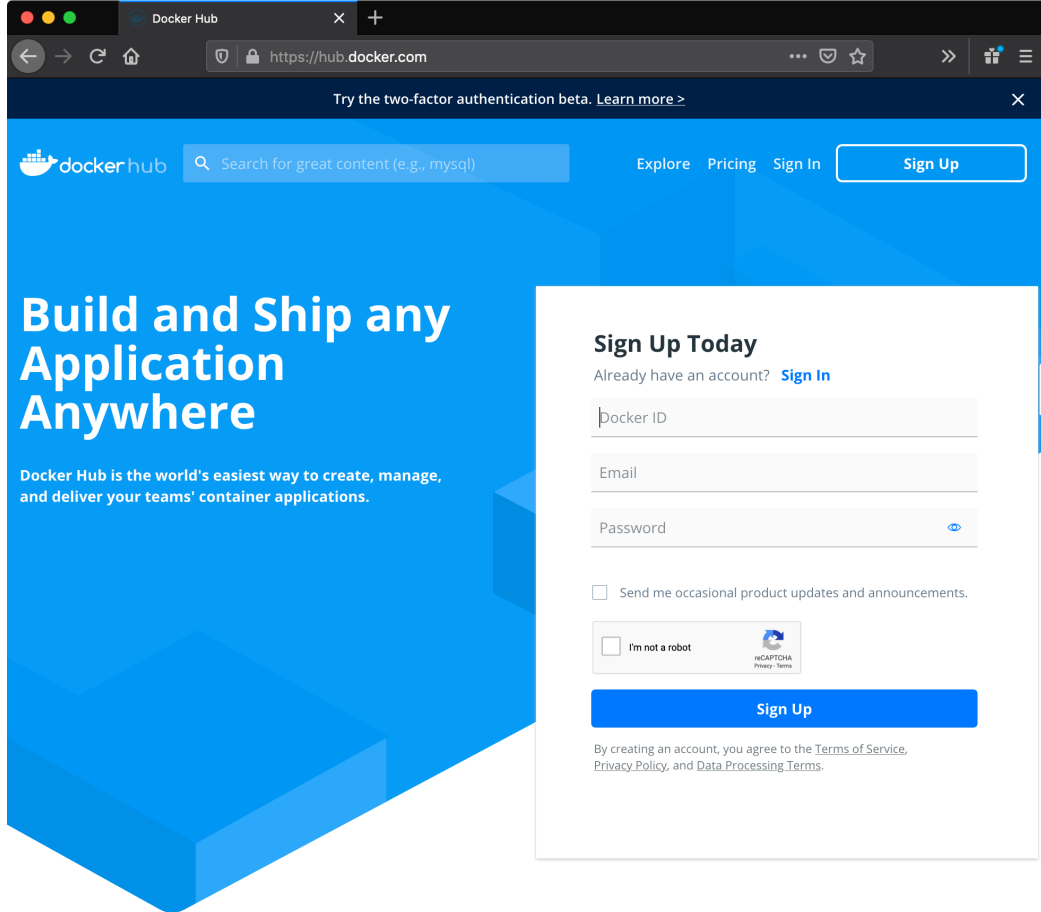


Virtual machines



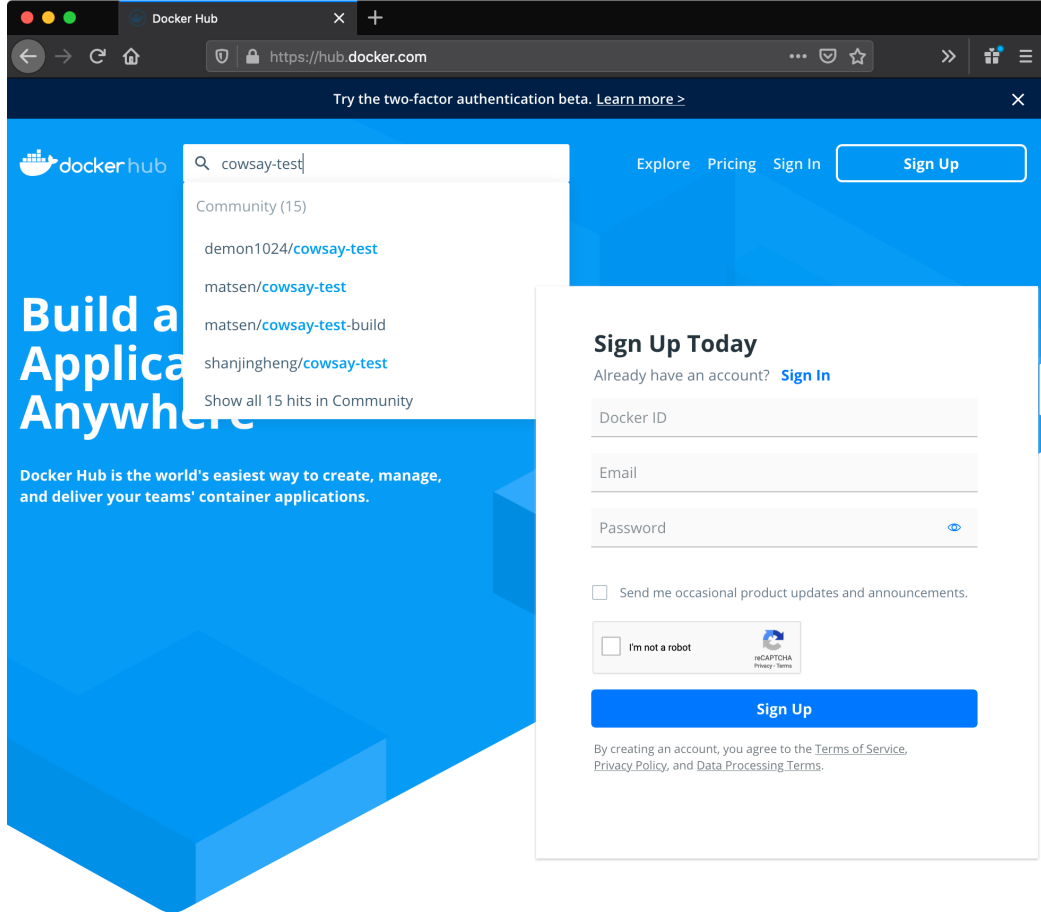
Containers

Docker Hub



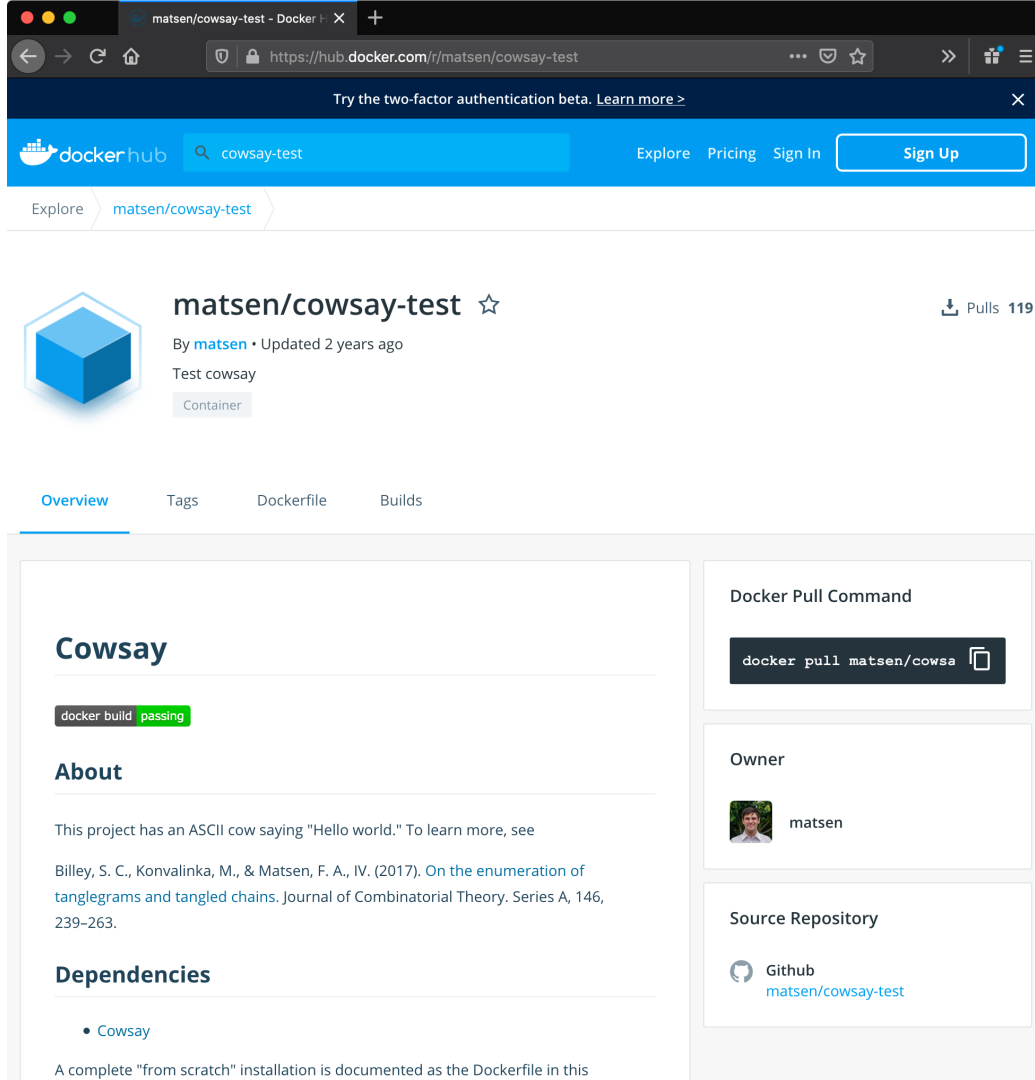
Docker Hub is the world's largest
library and community for container images

Search



Docker Hub is the world's largest
library and community for container images

matsen/ cowsay-test



The screenshot shows a web browser window displaying the Docker Hub page for the `matsen/cowsay-test` image. The browser's address bar shows the URL `https://hub.docker.com/r/matsen/cowsay-test`. The Docker Hub header is visible with the Docker logo, a search bar containing `cowsay-test`, and links for `Explore`, `Pricing`, `Sign In`, and `Sign Up`. Below the header, the page title is `matsen/cowsay-test` with a star icon and a pull count of 119. The image is described as "Test cowsay" and is a "Container" type. The `Overview` tab is selected, showing the image name `Cowsay` and a status of `docker build passing`. The `About` section explains that the project has an ASCII cow saying "Hello world." and references a paper by Billey, S. C., Konvalinka, M., & Matsen, F. A., IV. (2017). The `Dependencies` section lists `Cowsay` as a dependency. On the right side, the `Docker Pull Command` is shown as `docker pull matsen/cowsa`, the `Owner` is `matsen`, and the `Source Repository` is `GitHub matsen/cowsay-test`.

matsen/cowsay-test

Try the two-factor authentication beta. [Learn more >](#)

dockerhub cowsay-test Explore Pricing Sign In Sign Up

Explore matsen/cowsay-test

 **matsen/cowsay-test** ☆ 119 Pulls

By [matsen](#) • Updated 2 years ago

Test cowsay

Container

Overview Tags Dockerfile Builds

Cowsay

docker build passing

About

This project has an ASCII cow saying "Hello world." To learn more, see

Billey, S. C., Konvalinka, M., & Matsen, F. A., IV. (2017). [On the enumeration of tanglegrams and tangled chains](#). Journal of Combinatorial Theory. Series A, 146, 239–263.

Dependencies

- Cowsay

A complete "from scratch" installation is documented as the Dockerfile in this

Docker Pull Command

```
docker pull matsen/cowsa
```

Owner

 [matsen](#)

Source Repository

 [Github matsen/cowsay-test](#)



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Dockerfile

The screenshot shows a web browser displaying the Docker Hub page for the repository 'matsen/cowsay-test'. The browser's address bar shows the URL 'https://hub.docker.com/r/matsen/cowsay-test/dockerfile'. The page header includes the Docker Hub logo, a search bar with 'cowsay-test', and links for 'Explore', 'Pricing', 'Sign In', and 'Sign Up'. Below the header, the repository name 'matsen/cowsay-test' is displayed with a star icon and 'Pulls 119'. The repository is described as 'Test cowsay' and is a 'Container'. The 'Dockerfile' tab is selected, showing the following content:

```
FROM ubuntu:trusty

RUN apt-get update -q && \
    apt-get install -y -q --no-install-recommends \
        cowsay

RUN ln -s /usr/games/cowsay /usr/bin

COPY . /test-cowsay
WORKDIR /test-cowsay
CMD ./test.sh
```

On the right side, the 'Source Repository' section shows a GitHub icon and the text 'Github matsen/cowsay-test'.



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Ubuntu

The screenshot shows the Docker Hub page for the Ubuntu official image. The page includes the Docker Hub logo, a search bar with 'ubuntu' entered, and navigation links for Explore, Pricing, Sign In, and Sign Up. The Ubuntu logo and name are prominently displayed, along with the text 'Docker Official Images' and a description: 'Ubuntu is a Debian-based Linux operating system based on free software.' Below this, there are filters for Container, Linux, IBM Z, PowerPC 64 LE, x86-64, 386, ARM 64, and ARM. The 'Description' tab is selected, showing a section for 'Supported tags and respective Dockerfile links' with a list of tags: 18.04, bionic-20200311, bionic, latest, 19.10, eoan-20200313, eoan, rolling, 20.04, focal-20200319, focal, devel, 14.04, trusty-20191217, trusty, and 16.04, xenial-20200212, xenial. A layered architecture diagram on the right shows three application layers (App 1, App 2, App 3) on top of Bins/Libs, which sit on the Docker Engine, then the Operating System, and finally the Infrastructure layer.

ubuntu - Docker Hub

Try the two-factor authentication beta. [Learn more >](#)

dockerhub ubuntu Explore Pricing Sign In Sign Up

Explore ubuntu

ubuntu ☆
Docker Official Images
Ubuntu is a Debian-based Linux operating system based on free software.

1B+

Container Linux IBM Z PowerPC 64 LE x86-64 386 ARM 64 ARM

Base Images Operating Systems Official Image

Description Reviews Tags

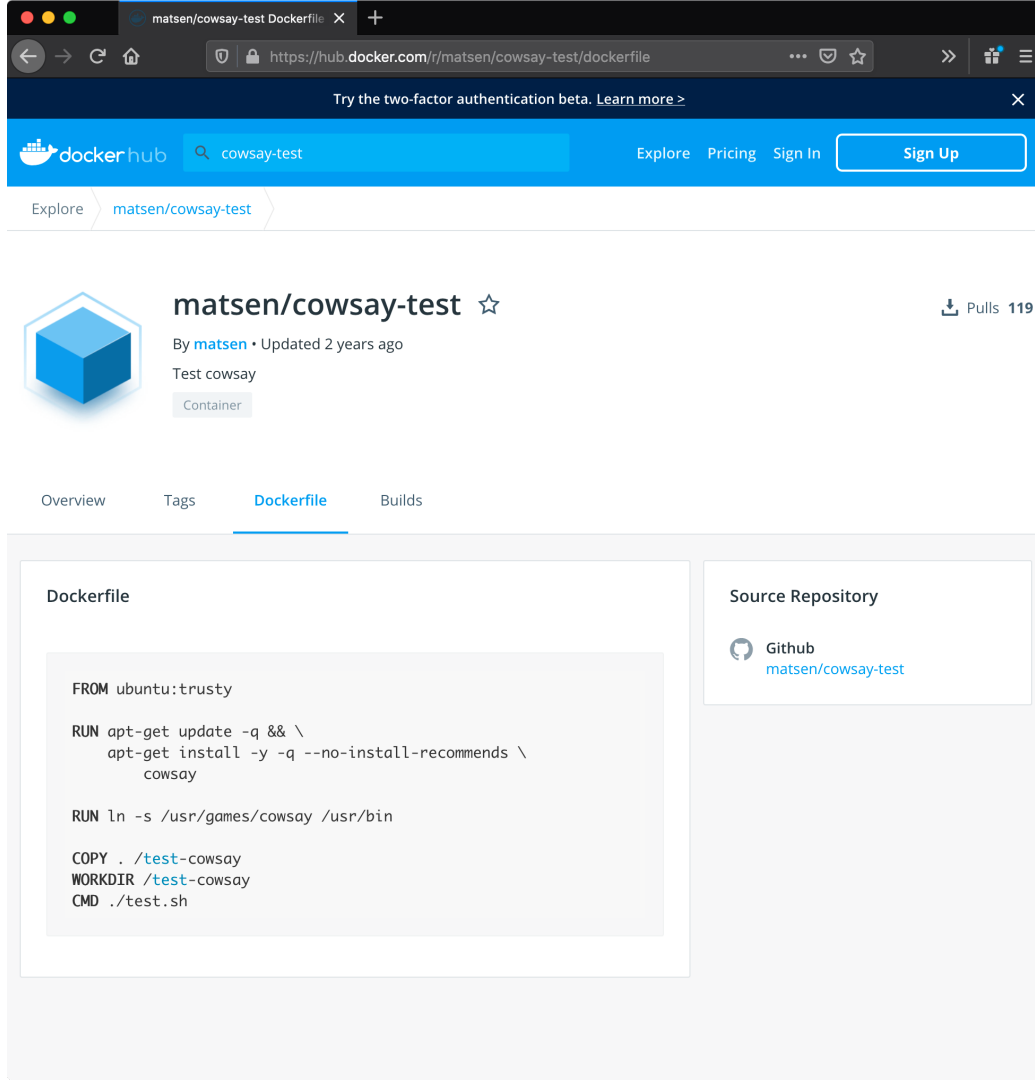
Supported tags and respective Dockerfile links

- 18.04, bionic-20200311, bionic, latest
- 19.10, eoan-20200313, eoan, rolling
- 20.04, focal-20200319, focal, devel
- 14.04, trusty-20191217, trusty
- 16.04, xenial-20200212, xenial

Quick reference

App 1 App 2 App 3
Bins/Libs Bins/Libs Bins/Libs
Docker Engine
Operating System
Infrastructure

Dockerfile



The screenshot shows a web browser displaying the Docker Hub page for the repository `matsen/cowsay-test`. The page includes a navigation bar with the Docker Hub logo, a search bar containing `cowsay-test`, and links for `Explore`, `Pricing`, `Sign In`, and `Sign Up`. Below the navigation bar, the repository name `matsen/cowsay-test` is prominently displayed, accompanied by a blue cube icon and a star icon. The page indicates it was updated 2 years ago and has 119 pulls. The `Dockerfile` tab is selected, showing the following content:

```
FROM ubuntu:trusty

RUN apt-get update -q && \
    apt-get install -y -q --no-install-recommends \
    cowsay

RUN ln -s /usr/games/cowsay /usr/bin

COPY . /test-cowsay
WORKDIR /test-cowsay
CMD ./test.sh
```

On the right side, the `Source Repository` section shows a link to the repository on GitHub.



Command line

```
root@81f14d0154d1: /test-cowsay#
```

```
> docker image pull matsen/cowsay-test
Using default tag: latest
latest: Pulling from matsen/cowsay-test
c2c80a08aa8c: Pull complete
6ace04d7a4a2: Pull complete
f03114bcfb25: Pull complete
99df43987812: Pull complete
9c646cd4d155: Pull complete
033d410e2c1e: Pull complete
a33cabb3e718: Pull complete
43fcbb1761798: Pull complete
Digest: sha256:957773822d7d42e0ceedcc9761b0cc58783c31904ddd18ff7c4589ed733e5db
Status: Downloaded newer image for matsen/cowsay-test:latest
docker.io/matsen/cowsay-test:latest
> docker images matsen/cowsay-test
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
matsen/cowsay-test   latest              2a52ce4f67e4       24 months ago      245MB
> docker run matsen/cowsay-test

< Hello world. >
-----
      \   ^__^
       (oo)\_______
        (__)\       )\/\
           ||----w |
           ||     ||

> docker run matsen/cowsay-test cowsay "Moo"

< Moo >
-----
      \   ^__^
       (oo)\_______
        (__)\       )\/\
           ||----w |
           ||     ||

> docker run -it matsen/cowsay-test /bin/bash
root@81f14d0154d1:/test-cowsay# ls
Dockerfile README.md docker.test.yml test.sh
root@81f14d0154d1:/test-cowsay# cat test.sh
cowsay "Hello world."
root@81f14d0154d1:/test-cowsay# ls /
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys test-cowsay tmp usr var
```



Let's do it ourselves

```
douglass-macbook-pro:~$ docker pull ubuntu:20.04
```

```
20.04: Pulling from library/ubuntu
```

```
345e3491a907: Pull complete
```

```
57671312ef6f: Pull complete
```

```
5e9250ddb7d0: Pull complete
```

```
Digest: sha256:cf31af331f38d1d7158470e095b132acd126a7180a54f263d386da88eb681d93
```

```
Status: Downloaded newer image for ubuntu:20.04
```

```
docker.io/library/ubuntu:20.04
```

```
douglass-macbook-pro:~$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
ubuntu	20.04	7e0aa2d69a15	4 days ago	72.7MB

```
douglass-macbook-pro:~$ docker run -it ubuntu:20.04 /bin/bash
```

```
root@d28664f23d4d:/# ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	07:23	pts/0	00:00:00	/bin/bash
root	9	1	0	07:24	pts/0	00:00:00	ps -ef

```
root@d28664f23d4d:/# apt-get update -q
```

```
...
```

```
root@d28664f23d4d:/# apt-get install -q cowsay
```

```
...
```

```
root@d28664f23d4d:/# /usr/games/cowsay "moo"
```

```
< moo >
```

```
-----
```

```
      \      ^      ^
       (oo)\_____)
        (_____)  )  /\
                ||----w |
                ||     ||
```



```

root@d28664f23d4d:/# exit
exit
douglass-macbook-pro:~$ docker run -it ubuntu:20.04 /bin/bash
root@e2c4cf63253c:/# /usr/games/cowsay "moo"
bash: /usr/games/cowsay: No such file or directory
root@e2c4cf63253c:/# exit
exit
douglass-macbook-pro:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
douglass-macbook-pro:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
e2c4cf63253c   ubuntu:20.04   "/bin/bash"            22 seconds ago Exited (127) 6 seconds ago             funny_gates
d28664f23d4d   ubuntu:20.04   "/bin/bash"            5 minutes ago  Exited (0) 24 seconds ago             nice_meninsky
douglass-macbook-pro:~$ docker rm e2c4cf63253c
e2c4cf63253c
douglass-macbook-pro:~$ docker rm nice_meninsky
nice_meninsky
douglass-macbook-pro:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES

```



Accessing host files

```
douglass-macbook-pro:~$ pwd
/Users/dpotter
douglass-macbook-pro:~$ docker run -it --rm ubuntu:20.04 /bin/bash
root@0053aa324313:/# ls /Users/dpotter
ls: cannot access '/Users/dpotter': No such file or directory
root@e83251495392:/# ls /home/
root@e83251495392:/# exit
douglass-macbook-pro:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
douglass-macbook-pro:~$ docker run -it --rm --mount type=bind,source=/Users/dpotter,target=/home
ubuntu:20.04 /bin/bash
root@ef3698e747bd:/# ls /home/
5tB4LJZv.json Pictures ...
root@ef3698e747bd:/# exit
```



CSCS (Sarus)

```
dpotter@daint102:~> module load sarus
dpotter@daint102:~> sarus pull matsen/cowsay-test
# image      : index.docker.io/matsen/cowsay-test:latest
# cache directory : "/scratch/snx3000/dpotter/.sarus/cache"
# temp directory  : "/tmp"
# images directory : "/scratch/snx3000/dpotter/.sarus/images"
> save image layers ...
> pulling      : sha256:9c646cd4d1557832b81ce6e956a18f791bdcc470290be8b23f153811a8d372f
> pulling      : sha256:c2c80a08aa8cdf0629269bab360b225d96a511b4c4ccb5f9c1d2abe01ff23d6f
> pulling      : sha256:99df439878124298f9d5577c286e59d7f012baed7cf8d69d104f5d87d727c28
> pulling      : sha256:f03114bcfb255c8eaa7fe33bbe409e7070993ee96152b9a04a7b6db0c3da5130
> pulling      : sha256:43fcb1761798a04a3cdd0d71f8ba069c398f36ccab6c13630ec7df989ae37ed9
> pulling      : sha256:6ace04d7a4a23236b2e4578bc607a21157cfd3203d0e26a3d1dfcbb2b918db6
> completed    : sha256:9c646cd4d1557832b81ce6e956a18f791bdcc470290be8b23f153811a8d372f
> pulling      : sha256:033d410e2c1e50d15d2a2ee59649347f0896769c35adc1ac5c4d6688c474ea5b
> completed    : sha256:f03114bcfb255c8eaa7fe33bbe409e7070993ee96152b9a04a7b6db0c3da5130
> completed    : sha256:99df439878124298f9d5577c286e59d7f012baed7cf8d69d104f5d87d727c28
> completed    : sha256:43fcb1761798a04a3cdd0d71f8ba069c398f36ccab6c13630ec7df989ae37ed9
> pulling      : sha256:a33cabb3e7188dee4fd0825d8afdb2faa9d3c0e8f16c57229448b277498b06b8
> completed    : sha256:6ace04d7a4a23236b2e4578bc607a21157cfd3203d0e26a3d1dfcbb2b918db6
> completed    : sha256:a33cabb3e7188dee4fd0825d8afdb2faa9d3c0e8f16c57229448b277498b06b8
> completed    : sha256:033d410e2c1e50d15d2a2ee59649347f0896769c35adc1ac5c4d6688c474ea5b
> completed    : sha256:c2c80a08aa8cdf0629269bab360b225d96a511b4c4ccb5f9c1d2abe01ff23d6f
> expanding image layers ...
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:c2c80a08aa8cdf0629269bab360b225d96a511b4c4ccb5f9c1d2abe01ff23d6f.tar"
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:6ace04d7a4a23236b2e4578bc607a21157cfd3203d0e26a3d1dfcbb2b918db6.tar"
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:f03114bcfb255c8eaa7fe33bbe409e7070993ee96152b9a04a7b6db0c3da5130.tar"
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:99df439878124298f9d5577c286e59d7f012baed7cf8d69d104f5d87d727c28.tar"
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:9c646cd4d1557832b81ce6e956a18f791bdcc470290be8b23f153811a8d372f.tar"
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:033d410e2c1e50d15d2a2ee59649347f0896769c35adc1ac5c4d6688c474ea5b.tar"
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:a33cabb3e7188dee4fd0825d8afdb2faa9d3c0e8f16c57229448b277498b06b8.tar"
> extracting    : "/scratch/snx3000/dpotter/.sarus/cache/sha256:43fcb1761798a04a3cdd0d71f8ba069c398f36ccab6c13630ec7df989ae37ed9.tar"
> make squashfs image: "/scratch/snx3000/dpotter/.sarus/images/index.docker.io/matsen/cowsay-test/latest.squashfs"
dpotter@daint102:~> sarus run matsen/cowsay-test
Detected glibc 2.19 (< 2.26) in the container. Replacing it with glibc 2.26 from the host. Please consider upgrading the container image to a distribution with glibc >= 2.26.
```

```
< Hello world. >
-----
      ^__^
      (oo)\_______
      (__)\       )\/\
          ||----w |
          ||     ||
```

```
dpotter@daint102:~> █
```




Omitted

- Details of building a Docker image
 - There are various tricks to optimize
 - Image layers for example
 - Multi-stage builds to reduce the size
 - You only need gcc to compile, not to run
- Orchestration
 - Create a pipeline and execute it
 - Kubernetes is one such system